EE Dept, IIT Bombay
Academic Year: 2025-2026, Semester I (Autumn)

## Course: MS101 Makerspace

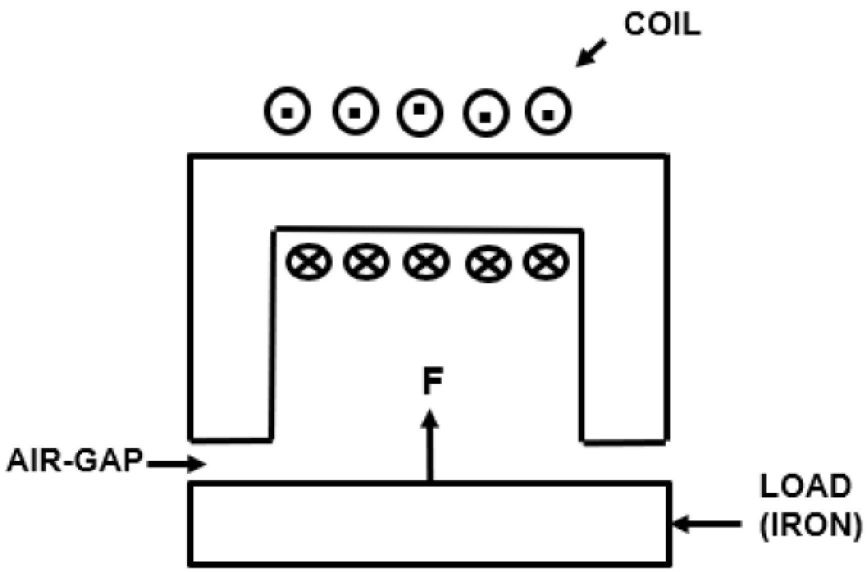
# EE Lecture 11
# Electromechanical Components

*Topics*

1. Electromagnet
2. Electromagnetic Relay
3. Relay Control
4. Solenoid
5. DC Motor
6. Brushless DC Motor
7. DC Servomotor

# 1. Electromagnet

- An electromechanical actuator, consisting of a magnetic core (soft iron or steel) and an excitation coil.
- Current flow in the coil results in magnetic field distribution with corresponding N and S poles.
- A high flux density (B) can be obtained in the air gap, subject to the core saturation. Energy stored in the air gap is used to carry out various functions (lifting material, closing/ opening contacts, etc.).
- Its main applications involve pulling an iron piece as a load. The pulling action is not affected by the current direction.




Tractive force across each air-gap: $f = \left(B^2 A\right)/(2\mu_0)$

$B$ = flux density in the air gap (assumed as uniform), proportional to the coil current.

$A$ = cross-sectional area of pole-face.

Total force: $F = 2f$.

For holding the load (mass $m$), $F > mg$.

# Electromagnet application example

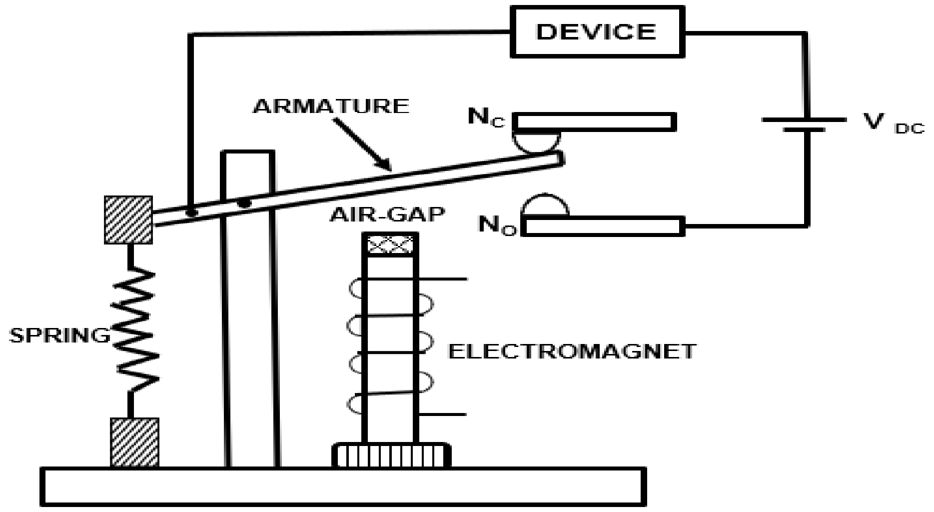
Relocation of steel pipes

# 2. Electromagnetic Relay

- A device for switching large voltages & currents using a relatively small current to control the excitation coil of an electromagnet. The controlling circuit is electrically isolated from the load circuits.
- Also known as electromechanical relay or EM relay.

*Most Common EM Relays:* Attracted Armature type.

- It consists of
  a) a moving part (armature) capable of making electrical connection with two contacts,
  b) an electromagnet with an excitation coil wound on a magnetic core, and
  c) a restraining spring.




- Relay operation depends on the coil ampere-turns, the air-gap between the armature and the core, and the restraining force of the spring.

Force on the armature: $F = KI_{rms}^2 - C$

$I$ = current through the excitation coil,

$K$ = proportionality constant,

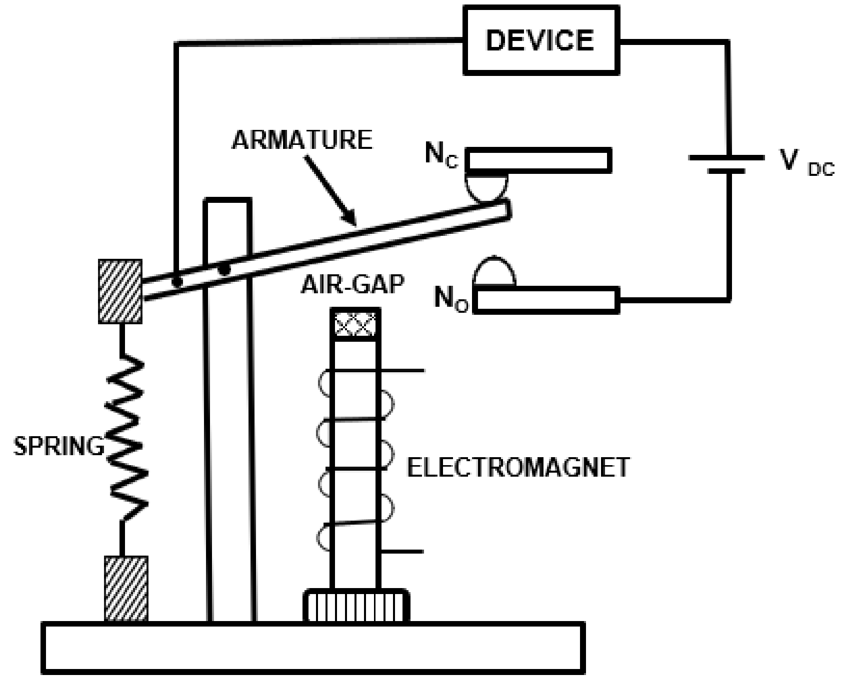
$C$ = force offered by the restraining spring.

# Attracted Armature Type Relay Operation

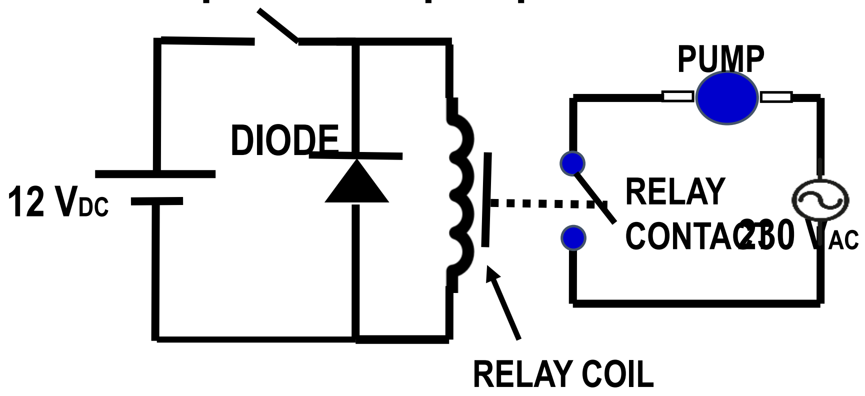
- A coil wound over a ferromagnetic material forms an electromagnet.
- Armature is the moving part of the magnetic circuit. It controls contacts and is restrained by a spring.

  *NC contact:* Normally Closed in un-excited mode.

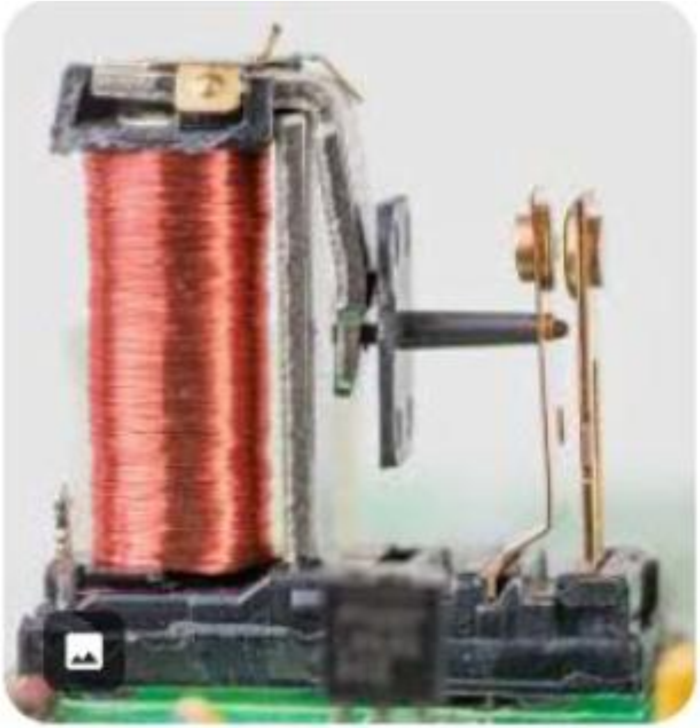
  *NO contact:* Normally Open in un-excited mode.
- When the coil is excited, the armature is attracted towards the NO contact. When the excitation current is turned off, the restraining force of the spring restores the armature to the NC contact.
- The armature movement between the two contacts makes and breaks the circuit ($V_{DC}$, DEVICE, Relay contacts).
- The excitation circuit controlling the relay coil current is electrically isolated from the load circuit connected to the relay contacts.



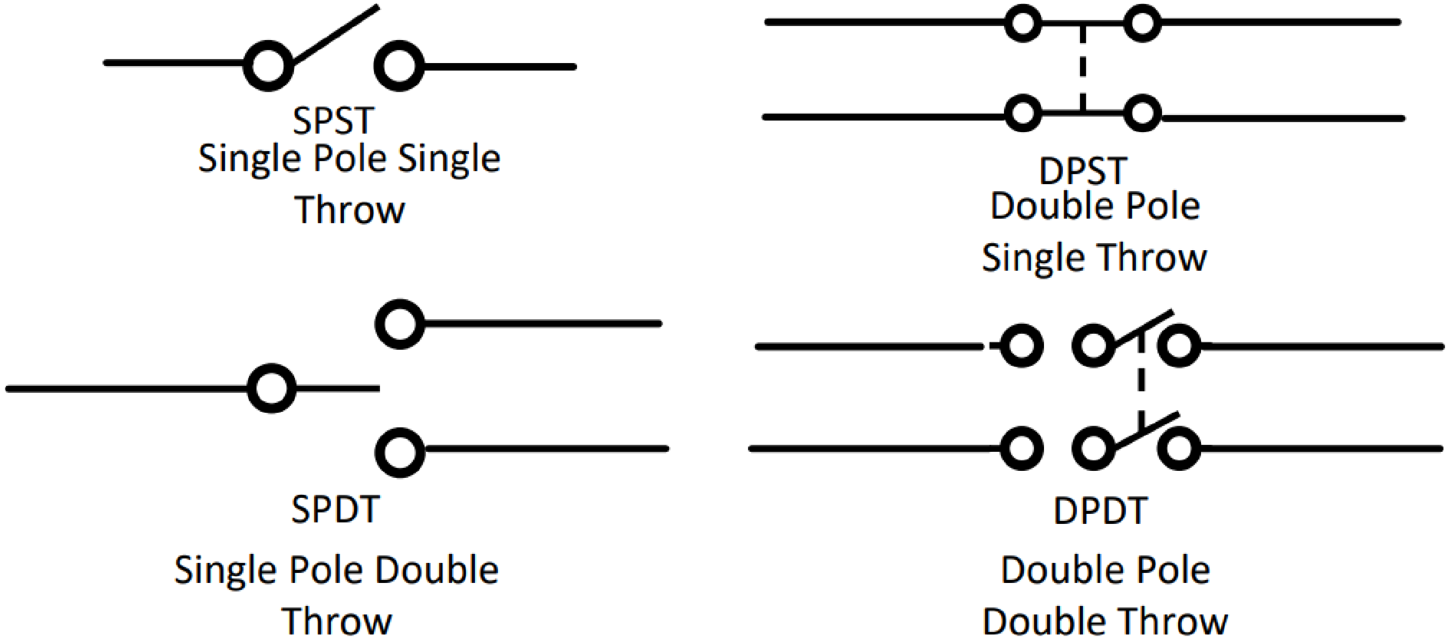

Example: Water pump ON-OFF control



# Relay Examples



## Relay Nomenclature




## Application examples

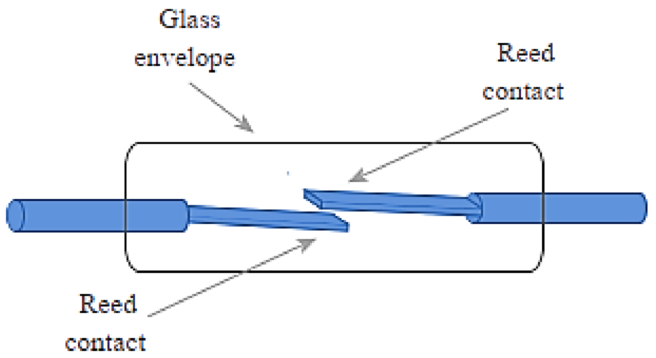
- SPST: On/off control of a load circuit by switching in one wire (e.g., AC mains phase wire).
- DPST: On/off control of a load circuit by simultaneous switching in both wires (e.g., AC mains phase & neutral).
- SPDT & DPDT: Complementary on/off control of two load circuits.

# Reed Relay

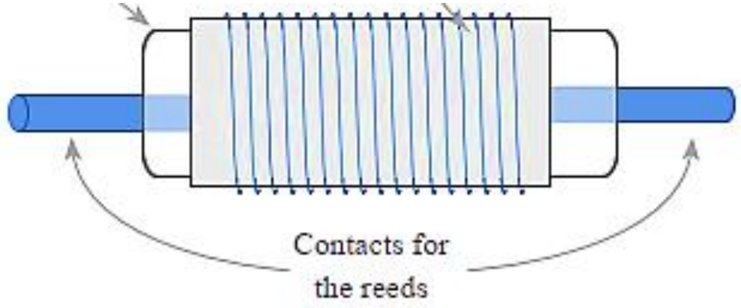
In this type of EM relay, the switch contacts are made of a ferromagnetic material, and the electromagnet acts directly on them without requiring an armature and spring. The contacts are sealed in a glass tube and protected from corrosion. The coil is placed around the tube.

- There may be multiple switches in the same tube.
- They can switch fast because they have small and lightweight moving parts.
- They require low operating power and have low contact capacitance.
- They are available in small packages.
- They have longer life, but lower current & voltage rating than the armature-based relays.

Reed switch inside the glass tube (reed: thin, wide, & flexible part, resembling the reed of a musical instrument)



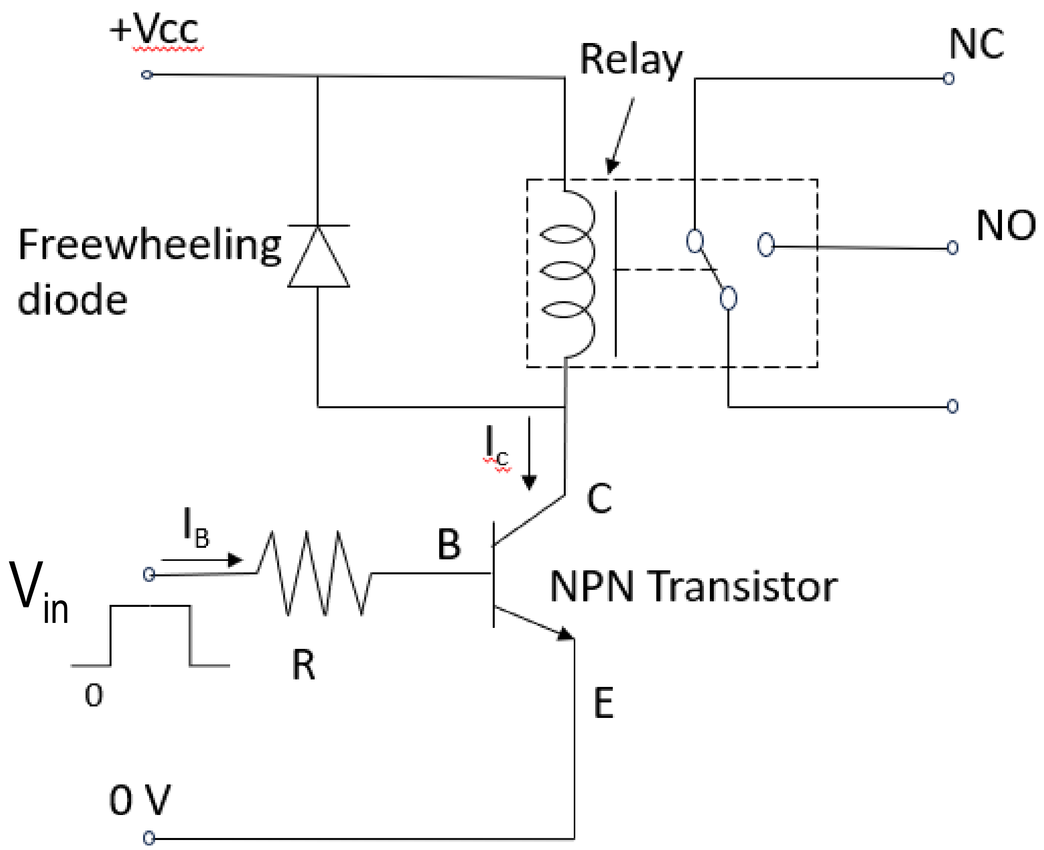

Reed relay (coil outside & reed contact inside the glass tube)




Packaged reed relays

# 3. Relay Control

***NPN transistor circuit to control the relay coil connected to the positive supply end (Vcc)***

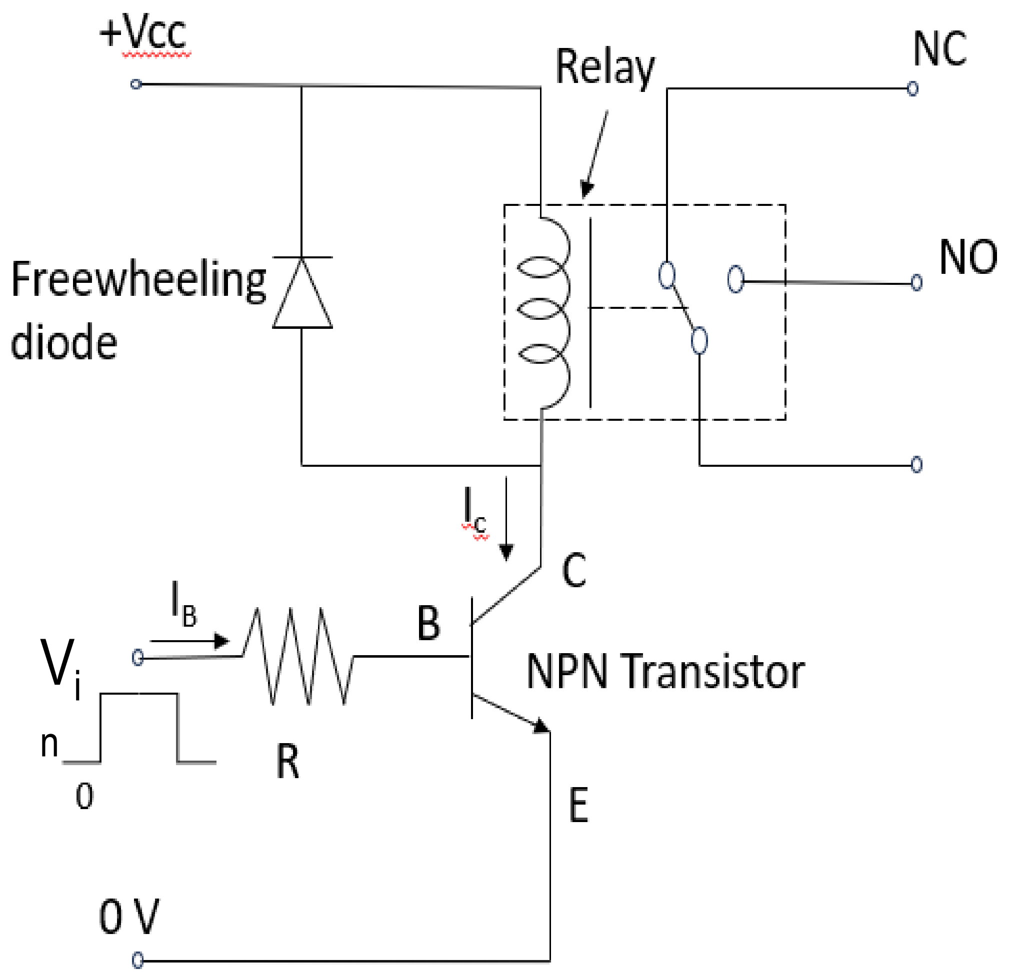



*On-state:* $V_{BESat} \approx 0.8$ V. $V_{CESat} \approx 0.2$ V.
Collector current: $I_C = (V_{cc} - V_{CESat}) / R_{coil}$
Current gain: $\beta = I_C / I_B$ .
For transistor in saturation mode, $I_B > I_C / \beta_{min}$.
Vin & $R$ should be such that

$$I_B = (V_{in} - V_{BESat}) / R > I_C / \beta_{min}.$$

*Off-state:* $V_{in} < V_\gamma$ , $V_\gamma = 0.5$ V. $I_B \approx 0$. $I_C \approx 0$.

- Voltage developed across the coil inductor: $V = L\, di/dt$
- When the transistor turns off, a sudden reduction in the coil current gives rise to a large reverse voltage at the transistor's collector terminal.
- The freewheeling diode avoids damage to the transistor. As the transistor turn-off, $I_c$ falls, $V_c$ increases, diode turns on and limits $V_c$ to $V_{cc}$+$V_D$. The coil current exponentially decays to zero (with time constant $L_{coil}/R_{coil}$), and $V_c$ settles at $V_{CC}$.




***Example***

$R_{coil}$ = 225 Ω.

$V_{CEsat}$ = 0.2 V. $V_{BEsat}$ = 0.8 V. $\beta_{min}$ = 35.

$V_{cc}$= 12 V. $V_{in}$(off) = 0 V. $V_{in}$(on) = 5 V.

Find *R* and the on-state coil current.

***On-state***

$I_C = (V_{cc} - V_{CEsat})/R_{coil}$

$=$(12− 0.2) / 225 $=$ 52.44 mA

$I_B > I_C/\beta_{min} =$ 52.44 / 35 = 1.50 mA

$V_{in} =$ 5 V.

$I_B$ = ($V_{in}$—$V_{BESat}$) / *R* = (5-0.8) / *R* = *4.2* / *R* >1.50 mA

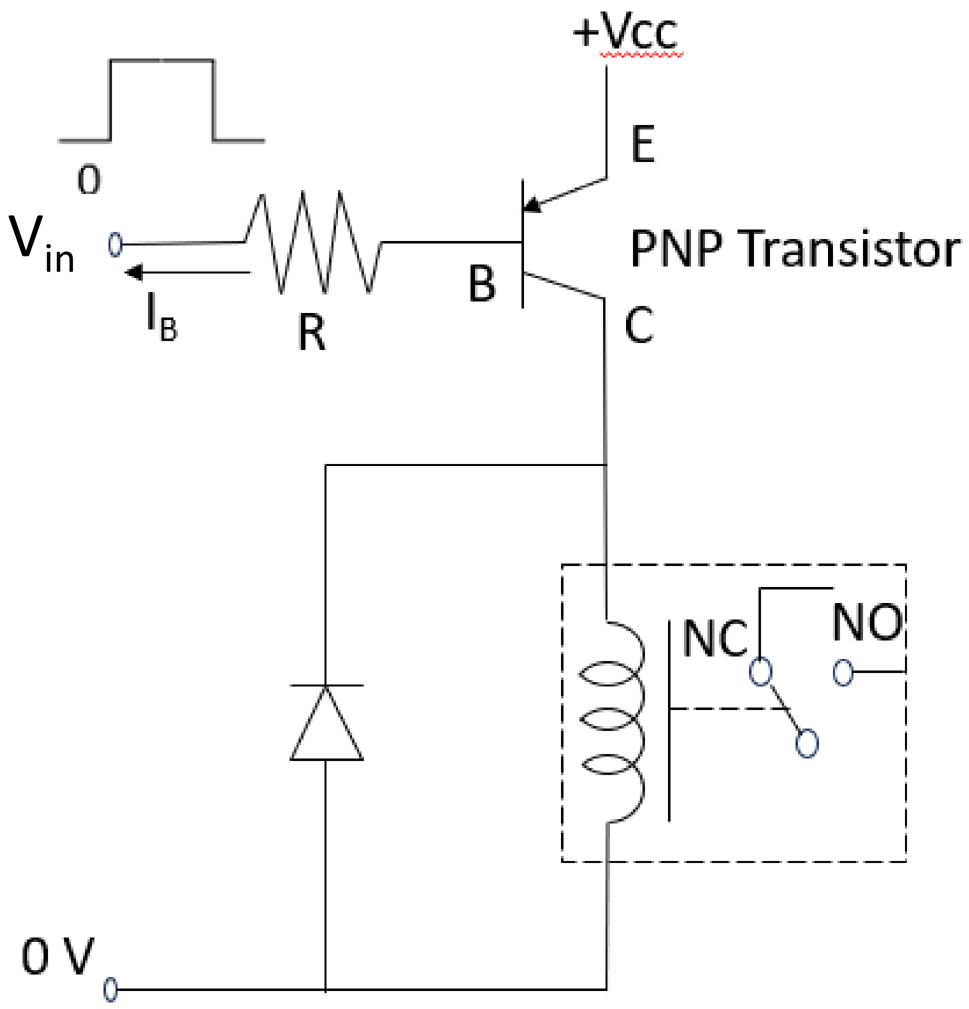
$\Rightarrow$ *R* < 4.2 / 1.50 kΩ = 2.8 kΩ.

Therefore we can use *R* = 2.7 kΩ.

***PNP transistor circuit to control the relay coil connected to the negative supply end (GND)***

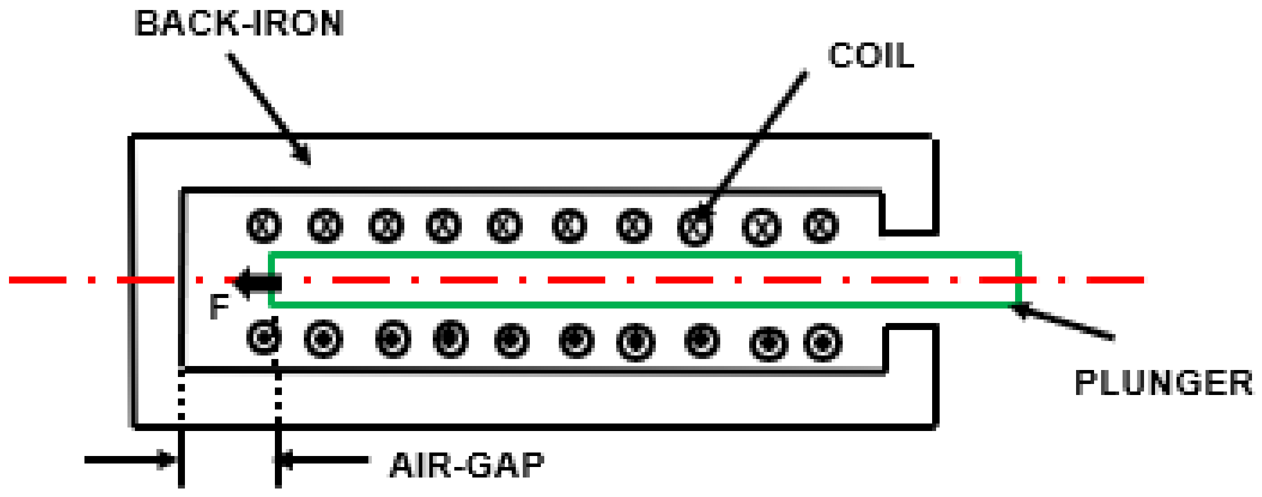
**The relay coil in both circuits is on the collector side of the switching transistor.**

- **PNP transistor circuit: one terminal of the relay is connected to GND.**
- **NPN transistor circuit: one terminal of the relay is connected to $\mathbf{V_{cc}}$.**



# 4. Solenoid

- An electromechanical actuator that uses electrical energy to cause mechanical movement.
- Normally used to
  a) push or pull a plunger (linear motion),
  b) realize rotatory motion (over an angle),
  c) open or close a valve.




- Typical Construction of a Linear Solenoid
  - A movable plunger or armature placed concentrically inside a coil.
  - Back-iron providing a low-reluctance magnetic path for flux.
- Coil current creates a magnetic field.
- Coil control circuit is same as for relays.

Force: $F = (B^2\, A)/(2\mu_0)$ $\qquad B \approx \mu_0 NI/l$

$N$ = number of turns. $I$ = coil current. $l$ = airgap length.
$A$ = plunger cross-sectional area.

$B$ = air-gap flux density.

# Solenoid Examples

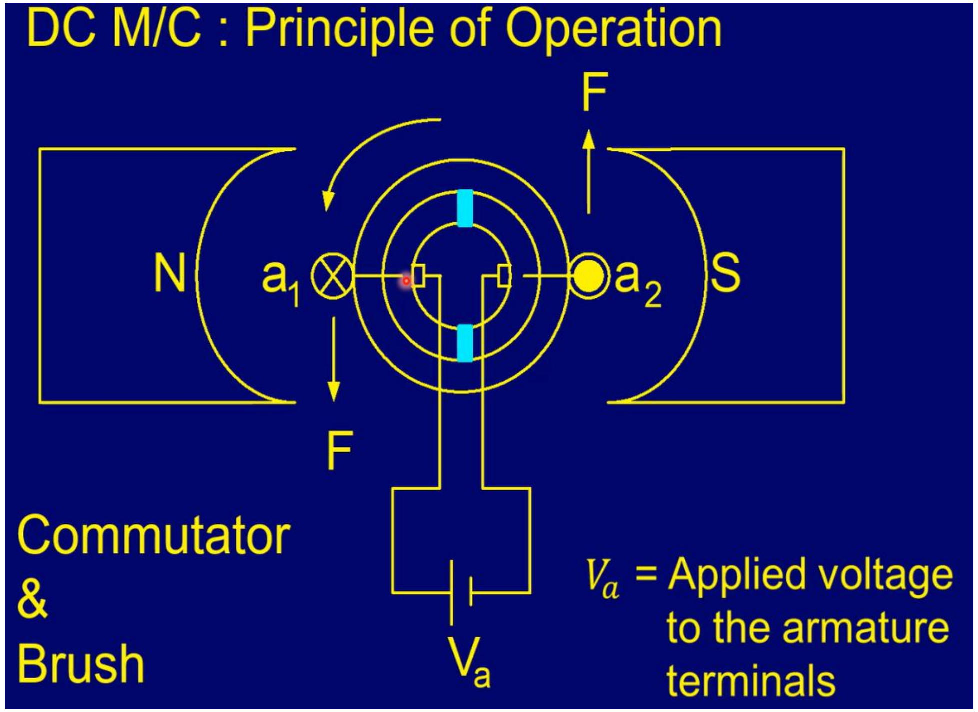
Linear motion solenoid with restoring spring

Bidirectional rotary solenoid

Solenoid-controlled fluid valves


# 5. DC Motor

**It is a DC-powered electromechanical device using interaction between the magnetic fields of a stator and a rotor (armature) for rotary motion of a load connected to its shaft.**




## *Conventional Brushed DC Motor*

- **Stator field is produced by its coils (electromagnets). Rotor coils have their two sides (a1 and a2 in the figure) pole-pitch apart under N and S poles.**
- **Rotor conductors carrying current and placed in the stator field experience force as per Fleming's Left Hand Rule (middle finger: current direction, index finger: stator field direction, thumb: force direction).**
- **As the rotor moves and the two coil sides exchange their positions, the directions of their currents are reversed by a commutator-brush arrangement to produce unidirectional torque.**

**Reference:**
https://docs.google.com/presentation/d/1s1Fsvb0lNAqGS9AAog9YmkMy3jo-zmj6/edit?usp=sharing&ouid=111902402391197382365&rtpof=true&sd=true
**To listen to the video, first download the shared PPT file, and then put it into slide show.**

- As the rotor coil moves in the field of the stator, a back electromotive force (e.m.f.) $E_b$ is induced in the rotor conductors. It is proportional to the stator flux ϕ & the angular speed ω. It opposes the rotor coil voltage $V_a$.

- The torque $T$ is proportional to rotor coil current $I_a$ and flux ϕ.

- $V_a = E_b + I_a R_a$.     $R_a$ = rotor coil resistance.

  $E_b = K_e \phi \omega$.

  $T = K_e \phi I_a$.     $K_e$ = machine constant..

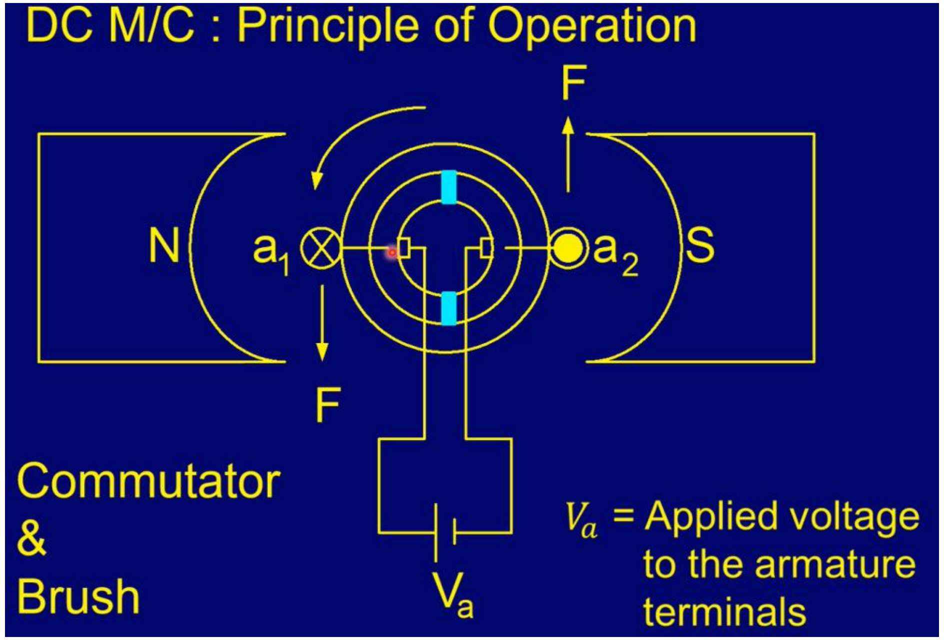
  Therefore,

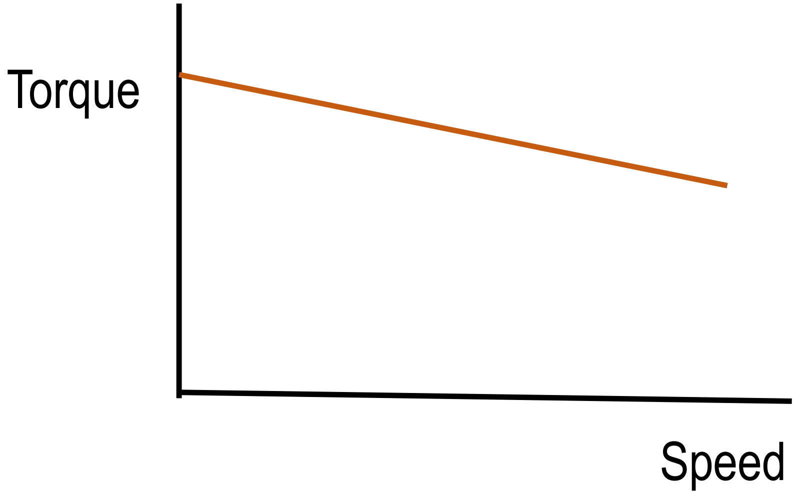
  $$\omega = E_b/(K_e \phi) = (V_a - I_a R_a)/(K_e \phi) = (V_a - (T/(K_e \phi) R_a)/(K_e \phi)$$

  $$= V_a/(K_e \phi) - R_a T/(K_e \phi)^2$$

- For a given $V_a$ and $\phi$ (proportional to stator coil current), speed $\omega$ decreases as the torque $T$ increases.

- For a given torque $T$, speed $\omega$ can be increased by increasing the rotor coil voltage $V_a$.



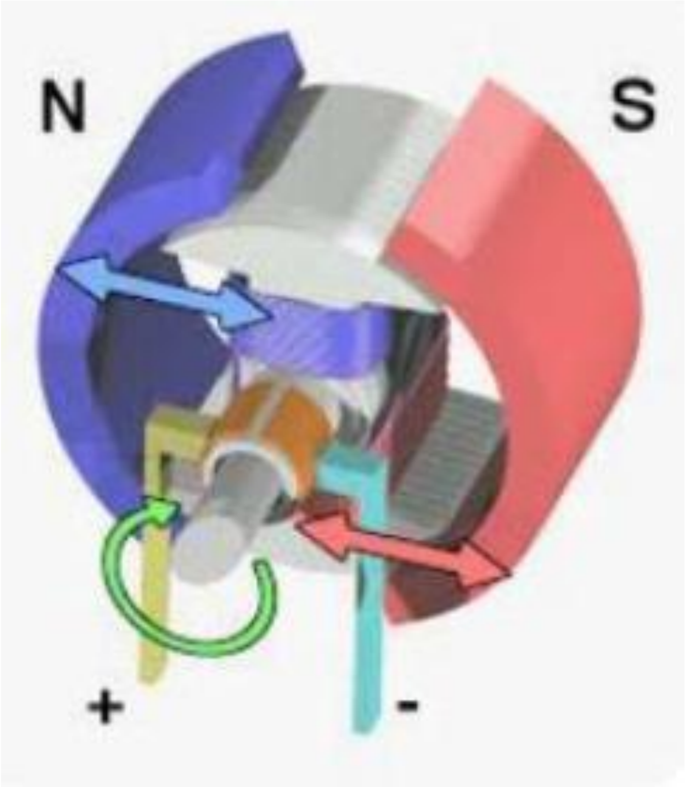
***Brushed Permanent Magnet DC (PMDC) motor***

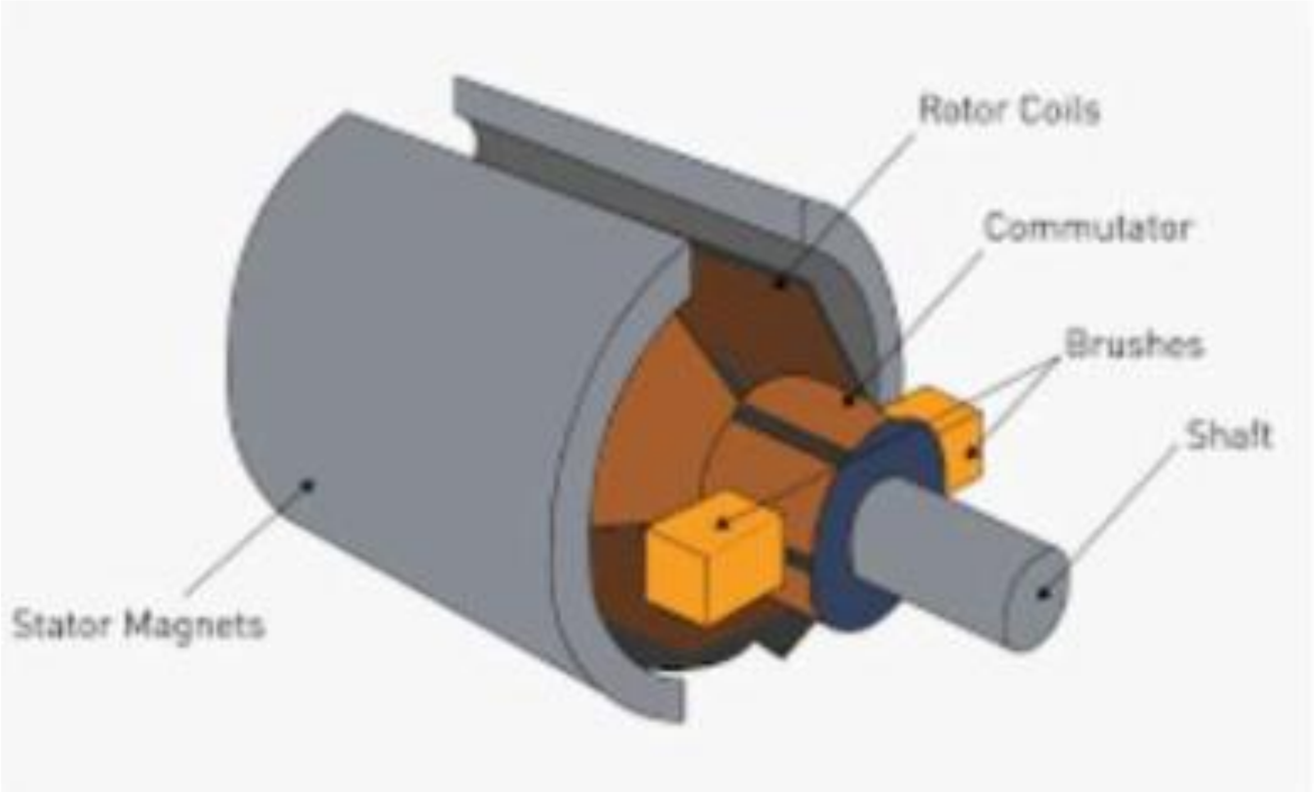
***The stator field is produced by permanent magnets. In these motors, only the rotor coil needs current.***

- **The direction of currents flowing in the rotor coils forming electromagnets are changed appropriately through the commutator-brush arrangement to produce unidirectional torque in accordance with the principle that like poles of the stator and the rotor repel and their opposite poles attract each other.**
- **During the current direction reversal in the rotor coils, the currents may get reversed before becoming zero. The coils have inductance, and interruption of non-zero currents in them results in large voltages between the commutator segment and the brushes, causing sparking between them. It results in large spikes in the applied DC voltage.**
- **To mitigate the effect of voltage spikes on the connected electronic components in the circuit, a suitable capacitor is connected across the motor terminals (as in Experiment 4).**
- **The motor speed can be varied by changing the applied DC voltage. The rotation direction can be controlled by changing the polarity of the applied voltage. These controls can be carried out using two SPDT switches or four SPST switches (H bridge).**

***Battery-operated (BO) motors are low voltage and geared PMDC motors, with the gears used for increasing the torque with a corresponding decrease in speed.***

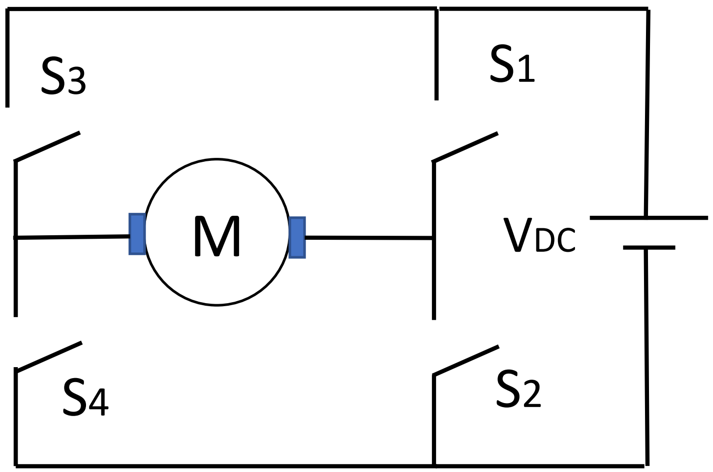
## Brushed Permanent Magnet DC (PMDC) motor: internal parts



## *DC (BO) Motor Speed & Direction Control*

- Speed can be controlled by changing the DC voltage. Rotation direction can be changed by changing the DC source polarity.
- Direction & speed can be controlled electronically using transistor switches connected as 'H-bridge' & freewheeling diodes to protect the transistors. These circuits are available as motor driver cards (e.g. L298). The speed is controlled by changing the average DC voltage by controlling the on/off duty cycle of the switches, known as pulse width modulation (PWM) control. For duty cycle $\alpha$, $V_{avg} = \alpha V_{DC}$.




*To be avoided:* S1 & S2 on, S3 & S4 on.

*Unpowered:* all switches off, only one on.

*Breaking:* S1 & S3 on, S2 & S4 on.

*Powered:* S1 & S4 on, S2 & S3 on.

### *Switch states & action*

| S1 | S2 | S3 | S4 | Action |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | Unpowered |
| 0 | 0 | 0 | 1 | Unpowered |
| 0 | 0 | 1 | 0 | Unpowered |
| 0 | 0 | 1 | 1 | $V_{DC}$ shorted |
| 0 | 1 | 0 | 0 | Unpowered |
| 0 | 1 | 0 | 1 | Breaking |
| 0 | 1 | 1 | 0 | Forward |
| 0 | 1 | 1 | 1 | $V_{DC}$ shorted |
| 1 | 0 | 0 | 0 | Unpowered |
| 1 | 0 | 0 | 1 | Reverse |
| 1 | 0 | 1 | 0 | Breaking |
| 1 | 0 | 1 | 1 | $V_{DC}$ shorted |
| 1 | 1 | 0 | 0 | $V_{DC}$ shorted |
| 1 | 1 | 0 | 1 | $V_{DC}$ shorted |
| 1 | 1 | 1 | 0 | $V_{DC}$ shorted |
| 1 | 1 | 1 | 1 | $V_{DC}$ shorted |

Given: inputs A (1: forward, 0: reverse) and B (1: move, 0: stop). Use logic gates to get four switch controls.

# 6. Brushless DC (BLDC) Motor

- ***Disadvantages of brushed DC motors***

**- High maintenance and short lifespan because of wear and tear of brushes.**

**- Low efficiency on account of friction between brushes and commutator segments (energy loss and heat).**

- ***BLDC motors***

**- The stator houses a star or delta-connected three-phase coils (acting as electromagnets). The rotor is of inner or outer type with permanent magnets.**

**- Electronic speed controllers commutate currents in the stator coils producing unidirectional torque. The currents in the stator coils are switched appropriately as the rotor rotates.**

**- In most BLDC motor applications requiring precise torque control, the rotor position is sensed by a Hall effect sensor, and accordingly the electronic switches of the converter are controlled to reverse currents at appropriate time instants.**

- ***Advantages of BLDC motors***

**Higher efficiency, lower maintenance, compactness, higher power-to-size ratio, and lower noise.**

# 7. DC servomotor

***These motors are used in applications requiring precise control of mechanical movements (e.g., angular position). They use negative feedback with the angular position as the control variable.***

***Components***

- **DC motor**
- **Position sensor (potentiometer connected to motor shaft)**
- **Feedback control (closed loop) with control electronics**
- **Gearbox (for obtaining high torque and low speed)**

***A typical DC servomotor has 3-pins:*** **(i) Supply (ii) Ground, & (iii) Control (PWM)**

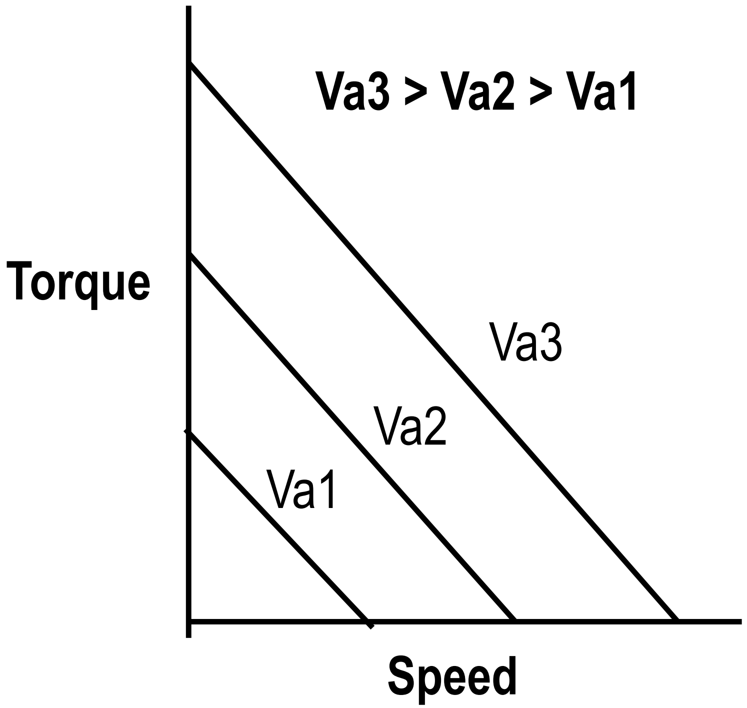
***Applications***

**Robotics, industrial manufacturing, machine tools, packaging, printing, automatic doors, steering systems of antennas, etc.**

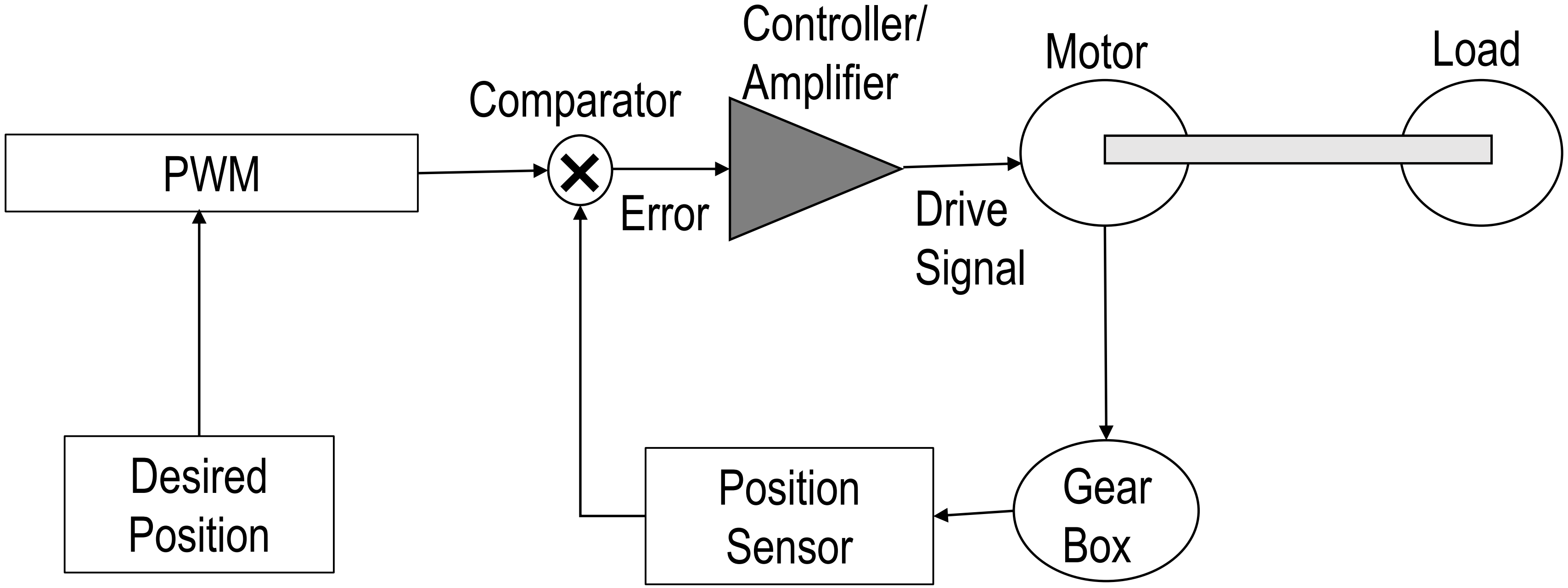
# *DC Servomotor Characteristics & Operation*

**DC motor: PMDC (brushed) or BLDC**

- **High torques at low speeds (achieved through a gearbox).**
- **Low inertia and fast response (motor construction with small diameter and long length).**
- **Not designed for continuous rotation (typically designed to operate over 0 to 180 degrees in less than one second).**
- **PWM control applied to the armature (applied voltage $V_a$ is changed with a suitable duty cycle).**
- **High-resistance armature winding: a large negative slope in torque-speed characteristics provides viscous damping.**



## *Servomotor Control*




***A servomotor is operated with closed-loop feedback with a position sensor (e.g., potentiometer).***

- **The error signal depends on the difference between the actual and desired positions.**
- **The drive signal to the motor should be such that the error is eventually reduced to zero, for which either proportional (P) control, proportional and integral (PI) control, or proportional, integral, and derivative (PID) control is used.**

**Reference:** https://www.elprocus.com/dc-servo-motor/

# *PID Control*

- P Control

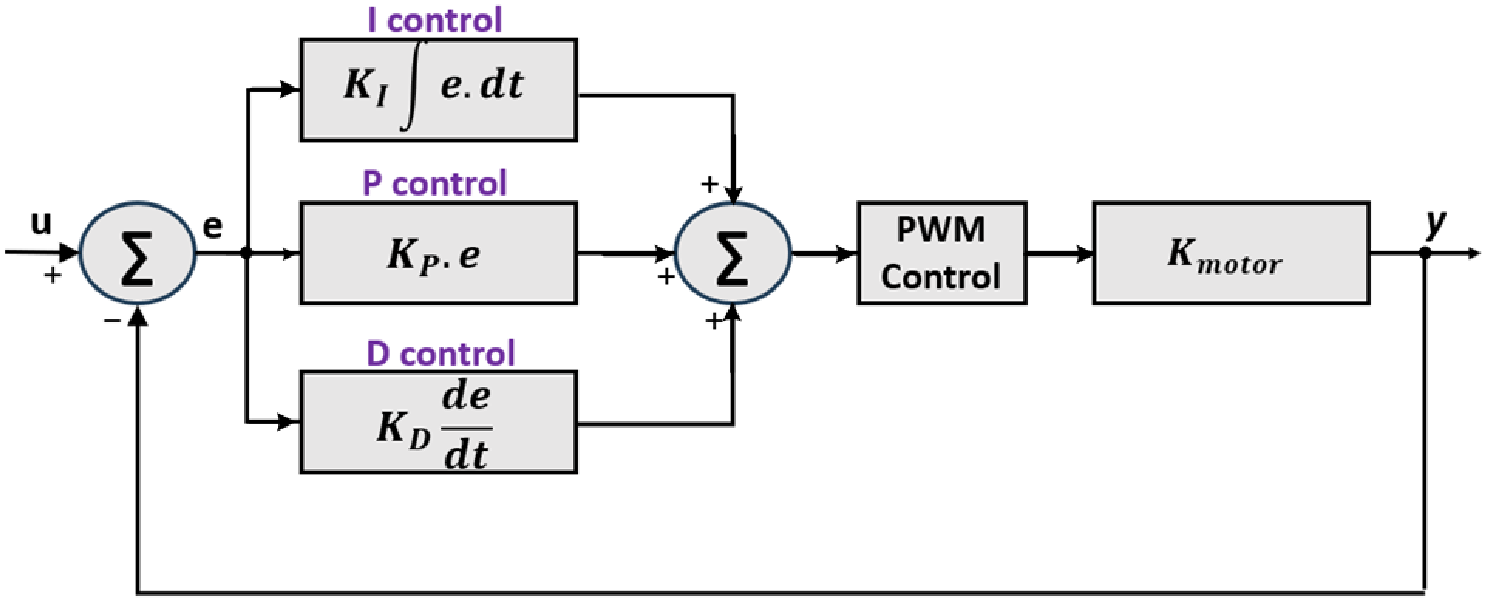
  $e = u - y$,

  Controller output: $K_p\, e$.

  A simple control approach, but it leads to a non-zero error.

  Steady-state offset =

  $$u/(1 + Kmotor\ .\ K_p)$$




- *PI Control:* Accumulated (integrated) error over some time is used to eliminate the offset (associated with P control). However, this control strategy results in an overshoot if the error accumulated in the initial period is high.

- *PID Control:* Derivative control block checks the rate of change of the error and dampens the overall gain by adding a term proportional to the negative of the rate. It helps to eliminate an overshot (if present).

- *Controller gains*: Depending on the system response (overshoots/ undershoots, fast/ slow response, etc.), the three controller gains need to be appropriately adjusted in a coordinated way to achieve the desired control characteristics.

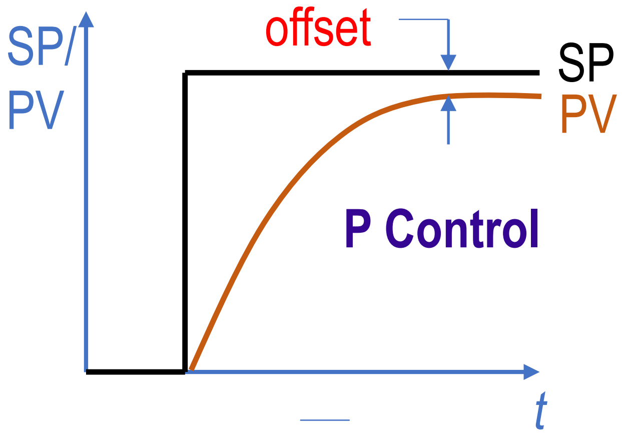
*PID Control Characteristics*

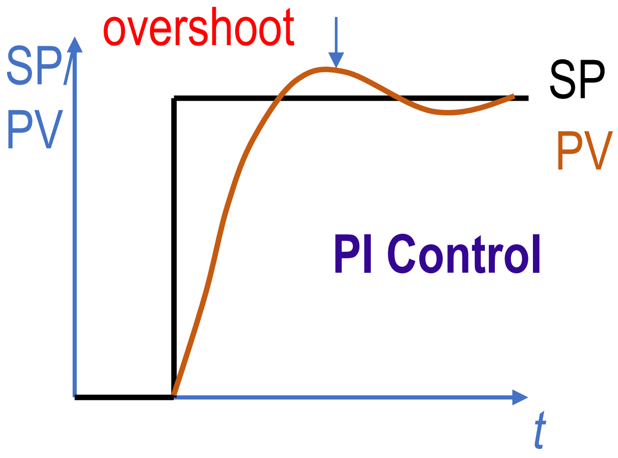
$u$ = Set Point (SP).

$y$ = Process Variable (PV), speed or position.

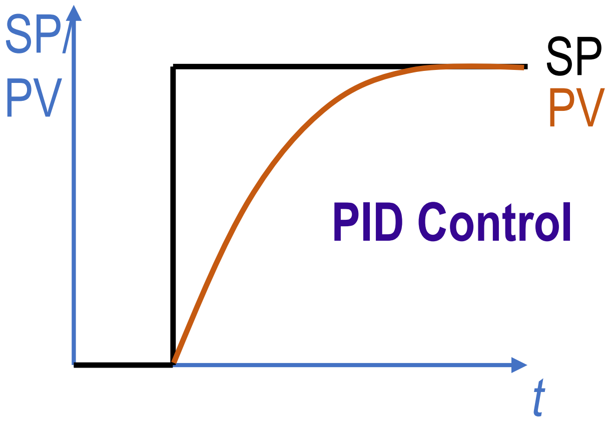
$e$ = error = SP − PV.

Controller gains: $K_P$, $K_I$, $K_D$




- P control can have a steady-state error.
- PI control can have an overshoot.
- In PID control, the three controller gains can be adjusted in a coordinated way to get the output free of steady-state error and overshoot.



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